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UNMANNED DELIVERY SYSTEM IN MARINE VESSEL AND CONTROL METHOD THEREFOR, AND MARINE VESSEL

Abstract

An unmanned delivery system in a marine vessel includes an acquisition unit to acquire destination information of a cargo to be delivered, an automatic navigation unit to cause a marine vessel to automatically navigate in an unmanned manner to a destination port indicated by the destination information, and a controller configured or programmed to issue an arrival notification and perform a fixed point holding of the marine vessel when the marine vessel arrives at the destination port.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of Japanese Patent Application No. 2024-31009, filed Mar. 1, 2024, which is hereby incorporated by reference herein in its entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates to unmanned delivery systems in marine vessels and control methods therefor, and marine vessels.

2. Description of the Related Art

[0003] In the field of automobiles, techniques for delivering a baggage by automatic driving have been developed (e.g., Japanese Patent No. 7211164). On the other hand, also in the field of marine vessels, if smooth automatic navigation becomes possible in the future, automatic delivery by unmanned navigation may be expected.

[0004] However, since marine vessel have unique circumstances different from those on land, it is required to develop automatic delivery techniques in consideration of these circumstances.

SUMMARY OF THE INVENTION

[0005] Example embodiments of the present invention provide unmanned delivery systems in marine vessels which enable unmanned delivery of cargoes via water.

[0006] According to an example embodiment of the present invention, an unmanned delivery system in a marine vessel includes an acquisition unit to acquire destination information of a cargo to be delivered, an automatic navigation unit to cause a marine vessel to automatically navigate in an unmanned manner to a destination port indicated by the destination information, and a controller configured or programmed to issue an arrival notification and perform a fixed point holding of the marine vessel when the marine vessel arrives at the destination port.

[0007] According to this configuration, destination information of a cargo to be delivered is acquired, a marine vessel is automatically navigated in an unmanned manner to a destination port indicated by the destination information, and an arrival notification is issued and a fixed point holding of the marine vessel is performed when the marine vessel arrives at the destination port.

[0008] The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the example embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a schematic side view of a marine vessel to which an unmanned delivery system is applied.

[0010] FIG. 2 is a block diagram of a main portion of the unmanned delivery system.

[0011] FIG. 3 is a conceptual diagram of unmanned delivery.

[0012] FIG. 4 is a diagram illustrating an example of a flow of a delivery control process in the unmanned delivery system.

[0013] FIG. 5 is a flowchart of the delivery control process.

DETAILED DESCRIPTION OF THE EXAMPLE EMBODIMENTS

[0014] Hereinafter, example embodiments of the present invention will be described with reference to the drawings.

[0015] FIG. 1 is a schematic side view of a marine vessel to which an unmanned delivery system according to an example embodiment of the present invention is applied. A marine vessel 100

includes a hull **101** and a marine propulsion device **102** mounted on the hull **101**. The marine propulsion device **102** is, for example, an outboard motor. Two or more marine propulsion devices **102** may be provided. The marine vessel **100** is communicably connected to a server **40** via a network **N**.

[0016] The marine propulsion device **102** is attached to the hull **101** via an attachment unit **114**. The marine propulsion device **102** includes a drive source (e.g., an engine) **103**. The drive source **103** may be an electric motor. The marine propulsion device **102** obtains a thrust to move the hull **101** by a propeller rotated by a driving force of the drive source **103**. The attachment unit **114** includes a swivel bracket, a clamp bracket, a steering shaft, and a tilt shaft.

[0017] The attachment unit **114** further includes a power trim and tilt mechanism (PTT mechanism). The PTT mechanism rotates the marine propulsion device **102** about the tilt shaft. This enables a change of an inclination angle (trim angle, tilt angle) of the marine propulsion device **102** with respect to the hull **101**. In addition, the marine propulsion device **102** is rotatable about the steering shaft with respect to the swivel bracket. During manual navigation, the marine propulsion device **102** turns left and right by operation of a steering wheel (not illustrated) such that the marine vessel **100** is steered.

[0018] A camera **24** is installed at a bow. The camera **24** may be attached, for example, directly to a bow rail or via a support. An imaging direction of the camera **24** is set forward or substantially forward. The camera **24** may be a stereo camera.

[0019] The marine vessel **100** is used to deliver a cargo. The marine vessel **100** includes a cargo room **106**. A loading lock **28** is provided on a door of the cargo room **106**, and when the loading lock **28** is unlocked, the door is opened to allow the cargo to be loaded (carried in) and taken out (carried out). A marine vessel door lock **27** is provided on a door **104** of the marine vessel **100**. An arrangement position of the door **104** is not limited. When the marine vessel door lock **27** is released, an operator can get on and off the marine vessel. Therefore, in order to load and take out the cargo, it is necessary to release the locks **27** and **28**. Note that it is not essential to provide the cargo room **106** and the loading lock **28**, and an entire floor surface of the hull **101** may be used as a cargo loading area.

[0020] As illustrated in FIG. **1**, a tag **109** (**109A**, **109B**) is attached to a cargo **108** (**108A**, **108B**) loaded in the cargo room **106** for delivery. The tag **109** records cargo specifying information. The cargo specifying information is correspondence information that associates the cargo **108** with its destination information. The cargo **108** and the destination are uniquely specified by the cargo specifying information. The information for specifying the cargo **108** and the destination is not limited to physical elements such as the tag **109**.

[0021] FIG. **2** is a block diagram of a main portion of the unmanned delivery system. The unmanned delivery system includes the marine vessel **100**, the server **40**, and one or more clients **50**.

[0022] In addition to the camera **24**, the marine vessel door lock **27**, and the loading lock **28** described above, the marine vessel **100** includes a navigation unit **21**, a sensor group **22**, a GNSS reception unit **23**, and a display unit **26**. In addition, the marine vessel **100** includes a CPU **11**, a ROM **12**, a RAM **13**, a memory **14**, a communication I/F (interface) **15**, a timer **16**, an input unit **17**, a weighing scale **18**, a waterline detection unit **19**, and a tag reading unit **20**.

[0023] The CPU **11** controls the entire marine vessel **100**. The ROM **12** or the memory **14** stores a control program. The memory **14** also stores chart information acquired from the server **40**. The CPU **11** implements various control processes by developing the control program stored in the ROM **12** or the like to the RAM **13** and executing the control program. The RAM **13** provides a work area when the CPU **11** executes the control program. The timer **16** counts time.

[0024] In addition to the drive source **103** (FIG. **1**), the navigation unit **21** includes elements necessary for navigation such as a steering, a remote controller, a throttle, a shift mechanism, a steering mechanism, and the like. In addition to a sensor to detect operation of the input unit **17**, the

sensor group **22** may include an azimuth sensor, an acceleration sensor, a speed sensor, an angular velocity sensor, an engine speed sensor, a shift position sensor, a millimeter wave radar, and the like. A detection result obtained by the sensor group **22** is sent to the CPU **11**.

[0025] The GNSS reception unit **23** periodically receives a global navigation satellite system (GNSS) signal from a GNSS satellite. As a result, the CPU **11** can acquire a current position of the marine vessel **100**.

[0026] The input unit **17** receives inputs of various settings, modes, and the like from a vessel operator of the marine vessel **100**. The communication I/F **15** can communicate with the server **40** via the network N, and also can communicate with an Electronic Control Unit (ECU) that controls the drive source **103** of the navigation unit **21** by a Controller Area Network (CAN) or the like.

[0027] The camera **24** includes a photographing unit that is fixed to the hull **101** and takes photographs or images of the surroundings of the marine vessel **100**. The display unit **26** displays various types of information. The captured image obtained by the camera **24** is sent to the CPU **11** and displayed on the display unit **26** in a manner according to a set mode.

[0028] The marine vessel door lock **27** and the loading lock **28** are changeable between a locked state and an unlocked state in accordance with an instruction from the CPU **11**. The weighing scale **18** is arranged on a floor surface of the cargo room **106**, measures a total weight of a cargo loaded in the cargo room **106**, and sends a measurement value to the CPU **11**. The waterline detection unit **19** detects a waterline around the marine vessel **100**. Although as the waterline detection unit **19**, for example, a detector such as an imaging device or an ultrasonic sensor can be used, the configuration thereof is not limited. The tag reading unit **20** reads information included in or on the tag **109** attached to the cargo.

[0029] In FIG. 2, an automatic navigation unit **29** includes elements indicated by the reference numerals **11** to **16** and **21** to **24**. In addition, an acquisition unit **30** includes the elements indicated by the reference numerals **11** to **20**. Note that it is not essential for each of the acquisition unit **30** and the automatic navigation unit **29** to include all of the elements illustrated in FIG. 2.

[0030] Although details will be described below, the acquisition unit **30** acquires destination information, cargo specifying information, and the like of a cargo to be delivered (loaded on the marine vessel **100**) in a delivery acceptance process **71** (FIG. 4).

[0031] The automatic navigation unit **29** automatically navigates the marine vessel **100** in an unmanned manner to a destination port indicated by the destination information. In consideration of an emergency or the like, a crew member may board the marine vessel **100** during delivery by the automatic navigation. As a basic method for achieving unmanned automatic navigation by the automatic navigation unit **29**, a known method can be used. The automatic navigation unit **29** achieves automatic navigation mainly by using the captured images obtained by the camera **24**, detection information obtained from the sensor group **22**, position information obtained from the GNSS reception unit **23**, and the like.

[0032] The server **40** includes a CPU **41**, a ROM **42**, a RAM **43**, a memory **44**, a communication I/F **45**, an input unit **47**, and a display unit **46**. The CPU **41** implements various control processes by developing a control program stored in the ROM **42** or in the memory **44** to the RAM **43** and executing the control program. The RAM **43** provides a work area when the CPU **41** executes the control program. The display unit **46** displays various types of information. The input unit **47** receives inputs of various settings and instructions from a user of the server **40**. The communication I/F **45** can communicate with the marine vessel **100** and the client **50** via the network N.

[0033] The client **50** includes a CPU **51**, a ROM **52**, a RAM **53**, a memory **54**, a communication I/F **55**, an input unit **57**, and a display unit **56**. The CPU **51** implements various control processes by developing a control program stored in the ROM **52** or the memory **54** to the RAM **53** and executing the control program. The RAM **53** provides a work area when the CPU **51** executes the control program. The display unit **56** displays various types of information. The input unit **57**

receives inputs of various settings and instructions from a user of the client **50**. The communication I/F **55** can communicate with the network N.

[0034] Note that the communication I/F **15**, **45**, and **55** may include a plurality of communication functions, and a method of the communication functions may be wired or wireless. In addition, any communication I/F may include a short-distance wireless communication function.

[0035] FIG. **3** is a conceptual diagram of unmanned delivery. The unmanned delivery is mainly achieved by the marine vessel **100** carrying a cargo from a base port **60** on land to an island (or land, hereafter island) **61** (**61A**, **61B**, **61C**) separated by the sea, and a recipient or the like unloading the cargo at a port (pier) of the island **61**. Although three islands **61** are illustrated in FIG. **3**, the number of the islands **61** may be one or four or more.

[0036] The client **50** (**50A**, **50B**, **50C**) includes a communication terminal device such as a smartphone owned by a user who receives a cargo on the island **61** (**61A**, **61B**, **61C**).

Communication terminal devices of users on the islands **61A**, **61B**, and **61C** are identified as the clients **50A**, **50B**, and **50C**, respectively. Note that one or more clients **50** may correspond to a single island **61**.

[0037] It is assumed that a predetermined application is installed in each client **50** in advance. As represented by the island **61A**, a mooring pillar **62** is provided at the pier of each island **61**. When the docked marine vessel **100** is anchored, the user of the client **50** couples the marine vessel **100** and the mooring pillar **62** by a mooring rope **63**. Hereinafter, the pier of the island **61** where a user receives a delivery may be referred to as a “destination port”.

[0038] In a case of delivering to a plurality of islands **61**, the server **40** determines a reasonable route for automatic navigation based on the information received from the CPU **11**. For example, in a case where a cargo is delivered to each of the islands **61A**, **61B**, and **61C**, the marine vessel is navigated in a route of the island **61A**.fwdarw.**61B**.fwdarw.**61C**.fwdarw. the base port **60**. A route in which an island **61** having no delivery scheduled may be excluded from the predetermined route. Therefore, an island **61** that does not correspond to any of the destination information of the plurality of loaded cargoes may be excluded from target ports of call. Note that, regardless of whether it corresponds to any of the destination information of the plurality of loaded cargoes or not, all the islands **61** may be visited in the predetermined route.

[0039] FIG. **4** is a diagram illustrating an example of a flow of a delivery control process in the unmanned delivery system. A representative flow of the delivery control process will be described with reference to FIGS. **3** and **4**. Information is exchanged between the marine vessel **100** and the client **50** via the network N and the server **40**. Note that the marine vessel **100** and the client **50** may directly communicate with each other if possible.

[0040] First, at the base port **60**, the delivery acceptance process **71** is executed before departure. A person who performs a series of tasks necessary for delivery at the base port **60** is referred to as a “sender” here. The sender may be a user at a cargo shipper or a staff member of a delivery company. In the delivery acceptance process **71**, the CPU **11** of the acquisition unit **30** acquires various types of information. Note that the method of acquiring various types of information is an example, and is not limited to the illustrated method.

[0041] The above various types of information can include weight and size of a cargo to be delivered, a desired delivery time zone, and the like, in addition to the destination information and the cargo specifying information of the cargo described above. The destination information, weight, size, and desired delivery time zone are input by a sender, for example. The sender attaches the tag **109** to the cargo. The CPU **11** (tag reading unit **20**) reads the cargo specifying information from the tag **109** attached to the cargo. The read cargo specifying information is stored in association with the destination information and the like. Note that reading of the tag **109** may be performed after loading.

[0042] Next, the CPU **11** executes a “loading guide” to guide the sender to a recommended loading area of the cargo in the cargo room **106** based on the weight of the cargo, the size of the cargo, or

the weight and size of the cargo. The display unit **26** may be used in the loading guide. In the loading guide, the CPU **11** determines an arrangement position of each cargo so that a weight distribution in the marine vessel **100** is as even as possible. Note that the weight and size of the cargo may be input as values obtained using the weighing scale, the camera, or the like, instead of being input by the sender. The sender loads the cargo according to the loading guide.

[0043] When it is determined that the loading of the cargo to be delivered is completed, the sender inputs information indicating loading completion. The information indicating the loading completion can be input by pressing a predetermined button, for example. The CPU **11** transmits all the acquired information including the cargo specifying information and the destination information to the server **40** on condition that departure permission information to permit departure has been generated.

[0044] In the present example embodiment, conditions for generating the departure permission information include that there has been an input of the loading completion by the sender and that a total weight of all the cargoes is appropriate. However, the condition that the total weight of all the cargoes is appropriate may not be included in the conditions for generating the departure permission information.

[0045] Whether the total weight is appropriate or not is determined as follows. The CPU **11** determines that the total weight is appropriate when a total weight of the cargoes loaded on the marine vessel **100** is within a first predetermined amount or when a total weight of the marine vessel **100** including the loaded cargoes is within a second predetermined amount. In other words, it is determined that the total weight is appropriate in a case where the total weight of all loads $\leq$ the first predetermined amount or “the total weight of all loads + a hull weight” $\leq$ the second predetermined amount. In determination of establishment of these expressions, the CPU **11** may determine the establishment of the above expression based on the total weight that is a total value of the individual loads input by the sender, or may determine the establishment of the above expression based on a position of the waterline detected by the waterline detection unit **19** or the total weight of the cargoes measured by the weighing scale **18**.

[0046] When the CPU **11** transmits all the information to the server **40**, the server **40** receives the information. The server **40** generates a password and associates the password with all the information. Furthermore, the server **40** determines a navigation route in a manner as described above, and returns a navigation start instruction including the determined route and password to the marine vessel **100**. The password may be issued by the marine vessel **100** and transmitted to the server **40** in association with the cargo specifying information.

[0047] When the marine vessel **100** receives the navigation start instruction from the server **40**, the CPU **11** (the automatic navigation unit **29**) starts automatic navigation toward the destination port. Therefore, the CPU **11** causes the marine vessel **100** to depart on condition that the shipper of the cargo has generated the departure permission information.

[0048] In parallel with the transmission of the navigation start instruction, the server **40** notifies the client **50** at the destination port as a delivery target of a delivery notice. The delivery notice includes, in addition to the cargo specifying information and the password, display information to display at least one of a list of cargoes to be unloaded at the destination port, an estimated time of arrival at the destination port, or an estimated time of departure from the destination port.

[0049] From this viewpoint, the CPU **41** of the server **40** defines and functions as a “transmission unit” that transmits the display information to the client **50** corresponding to the destination port. Note that the CPU **11** may be configured to play a role as the transmission unit by generating and transmitting the display information. Upon receiving the display information, the client **50** causes the display unit **56** thereof to display at least one of the list of cargoes to be unloaded, the estimated time of arrival at the destination port, or the estimated time of departure from the destination port. Note that the above display information may include the cargo specifying information and the information to display a password, and these may also be displayed on the display unit **56** of the

client **50**.

[0050] When the marine vessel **100** arrives at the destination port by automatic navigation, the CPU **11** as the controller executes an unloading process **72**. First, the CPU **11** transmits an arrival notification indicating that the marine vessel **100** has arrived at the destination port to the client **50** corresponding to the destination port. For example, the CPU **11** transmits an arrival notification indicating that the marine vessel **100** has arrived at the port of the island **61A** to the client **50A** corresponding to the port **61A**. The arrival notification includes the cargo specifying information.

[0051] In parallel with this, the CPU **11** starts a fixed point holding of the marine vessel **100**. The fixed point holding is control to keep the marine vessel **100** within a certain area. In the fixed point holding, for example, the CPU **11** calculates a thrust and a direction of each of the marine propulsion devices **102** required to eliminate a difference between the current position and a target position of the marine vessel **100**, and controls each of the marine propulsion devices **102** according to the calculation result.

[0052] Next, the CPU **11** waits for reception of “unloading preparation completion information” indicating that preparation to receive the cargo is completed. A person who performs a series of tasks necessary to pick up a cargo at a destination port is referred to as a “recipient” here. The recipient is usually an owner of the client **50** corresponding to the destination information. A recipient at a destination of the arrival notification (having received the arrival notification) inputs, to the client **50**, confirmation that the recipient intends to receive the cargo such that the unloading preparation completion information is issued. Alternatively, when a mooring sensor **64** (FIG. 3) detects that the user of the client **50** has coupled the marine vessel **100** and the mooring pillar **62** with the mooring rope **63**, the unloading preparation completion information is issued. Note that the unloading preparation completion information may be issued when either one of the input of the confirmation of intention of reception or the detection of the coupling with the mooring rope **63** has been performed. Alternatively, when both of them have been performed, the unloading preparation completion information may be issued.

[0053] The unloading preparation completion information is transmitted to the marine vessel **100**. In response to the acquisition of the unloading preparation completion information, the CPU **11** performs a “cargo unloading permission process” that enables unloading of the cargo. The cargo unloading permission process includes releasing the loading lock **28** of a door of the cargo room **106** or releasing the marine vessel door lock **27** of the door **104** for entering the marine vessel **100**. When only one of the locks is locked, only the locked one of the locks **27** and **28** needs to be unlocked.

[0054] After the cargo unloading permission process is performed, the recipient unloads the cargo addressed to the recipient himself/herself. At that time, the recipient specifies the cargo to be received by inputting the cargo specifying information, the password, or the like. After specifying the cargo to be received, the recipient starts unloading the cargo.

[0055] Note that when the user at the destination port has a cargo desired to be delivered to another port such as another island **61** or land, the client **50** transmits, at the destination port, a request for delivery to another port to the marine vessel **100**. When the marine vessel **100** receives the request for delivery to another port, the marine vessel **100** executes a process similar to the delivery acceptance process **71** performed at the base port **60**.

[0056] When the unloading work is completed, the recipient inputs, to the client **50**, information indicating completion of the unloading work. As a result, the departure permission information for permitting departure from the destination port is generated. The generated departure permission information is transmitted to the server **40**. When receiving the departure permission information, the server **40** determines a next navigation route and transmits the navigation start instruction including the navigation route to the marine vessel **100**. Note that, in a case where a new delivery acceptance process is performed, the initially determined navigation route may be changed, and contents of the delivery notice such as the estimated time of arrival at the destination port may also

be revised.

[0057] When receiving the departure permission information, the CPU **11** of the automatic navigation unit **29** causes the marine vessel **100** to depart. Note that, in a case where a predetermined period of time has elapsed after starting the fixed point holding at the destination port, the CPU **11** may cause the marine vessel **100** to depart even if the departure permission information is not generated. This is to reduce or prevent a delay in arrival at a subsequent destination port even when the recipient does not perform the reception at the destination port.

[0058] Therefore, the CPU **11** of the automatic navigation unit **29** normally causes the marine vessel **100** to depart on condition that the information indicating completion of the unloading work has been input to the client **50** at a destination of an arrival notification. However, even when the information indicating the completion of the unloading work is not input to the client **50**, the marine vessel **100** is caused to depart in a case where the predetermined period of time has elapsed after the fixed point holding is started.

[0059] The CPU **11** automatically navigates the marine vessel **100** to a next destination port, and executes the unloading process **72** when arriving at the next destination port. After repeating the unloading process **72** at each destination port and completing the unloading process **72** at all the destination ports, the CPU **11** controls the marine vessel **100** to return to the base port **60**.

[0060] Note that, in a case where the delivery acceptance process has been executed based on a request for delivery from the destination port to the other island **61**, the unloading process **72** corresponding thereto is executed in the other island **61**. In addition, in a case where the delivery acceptance process is executed based on a request for delivery from the destination port to land, after returning to the base port **60**, the work of unloading the loaded cargo and the process of delivery to the destination are performed.

[0061] FIG. **5** is a flowchart of the delivery control process. This process is achieved by the CPU **11** developing the program stored in the ROM **12** or the like to the RAM **13** and executing the program. This process is started in response to reception of a delivery acceptance start instruction. This process corresponds to the process executed by the marine vessel **100** in the flow of control illustrated in FIG. **4**. In particular, steps **S101** to **S103** are included in the delivery acceptance process **71**, and steps **S107** to **S109** are included in the unloading process **72**.

[0062] In step **S101**, as described with reference to FIG. **4**, the CPU **11** acquires various types of information such as the cargo destination information, the cargo specifying information, the weight and size of the cargo, and the desired delivery time zone, and executes the loading guide. In step **S102**, the CPU **11** determines whether departure is possible or not based on whether the departure permission information has been generated or not. As described above, in the present example embodiment, when there has been an input of loading completion by the sender and the total weight of all the cargoes is appropriate, it is determined that departure is possible. The CPU **11** repeats the process of steps **S101** and **S102** until the departure becomes possible, and transmits all the acquired information to the server **40** in step **S103** when the departure becomes possible.

[0063] In step **S104**, the CPU **11** waits until receiving the navigation start instruction from the server **40**, and when receiving the navigation start instruction, in step **S105**, the CPU starts automatic navigation in an unmanned manner toward a destination port indicated by the destination information.

[0064] In step **S106**, the CPU **11** continues the automatic navigation until the marine vessel **100** arrives at the destination port, and proceeds to step **S107** when the marine vessel **100** arrives at the destination port. In step **S107**, the CPU **11** transmits the arrival notification to the client **50** corresponding to the destination information at the arrived destination port, and starts the fixed point holding of the marine vessel **100**. Note that the transmission timing of the arrival notification and the start timing of the fixed point holding may not be the same, and neither which timing comes first nor an interval between both the timings matters.

[0065] In step **S108**, the CPU **11** determines whether unloading preparation is completed or not

based on whether the above-described unloading preparation completion information has been received or not. The CPU **11** stands by until the unloading preparation is completed, and when the unloading preparation is completed, the CPU **11** executes the cargo unloading permission process in step **S109**.

[0066] In step **S110**, the CPU **11** executes the remaining processes. In the remaining processes, for example, the CPU **11** executes the delivery acceptance process when the client **50** has made a request for delivery to another port.

[0067] In step **S111**, the CPU **11** waits until at least either of reception of the departure permission information from the server **40** or a lapse of the predetermined period of time after the fixed point holding is started occurs. Then, when either one of these occurs, the CPU **11** proceeds the process to step **S112**, and resumes the automatic navigation of the marine vessel **100** and causes the marine vessel to depart. Thereafter, the CPU **11** returns to step **S106**. Note that, in the remaining process in step **S110**, in a case where the marine vessel returns to the base port **60**, this process is ended, and a necessary unloading process is performed. Note that, in a case where an emergency occurs during this process, such a process may be executed as a remote control, a forcible returning to the base port **60**, or a temporary stopping at the nearest island **61**.

[0068] Note that, in the present example embodiment, a portion of the processes described as being executed by the CPU **11** may be executed by the server **40**, or by the CPU **11** and the server **40** in cooperation. Alternatively, the CPU **11** may execute a portion of the processes assumed to be executed by the server **40**, or the CPU **11** and the server **40** may execute the portion of the processes in cooperation.

[0069] Note that example embodiments of the present invention are also applicable to jet boats in addition to various marine vessels propelled by outboard motors, inboard motors, or inboard/outboard motors.

[0070] Although example embodiments of the present invention have been described in detail above, the present invention is not limited to these specific example embodiments, and various modifications without departing from the gist of the present invention are also included in the present invention.

[0071] Example embodiments of the present invention may also be processes in each of which a program executing one or more functions of the above-described example embodiments is supplied to a system or a device via a network or a non-transitory storage medium, and one or more processors of a computer of the system or the device reads and executes the program. The above program and the non-transitory storage medium storing the above program are also example embodiments of the present invention. Example embodiments of the present invention can also be implemented by circuits (e.g., ASIC) that implement one or more functions.

[0072] While example embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

Claims

1. An unmanned delivery system in a marine vessel, the unmanned delivery system comprising: an acquisition unit to acquire destination information of a cargo to be delivered; an automatic navigation unit to cause a marine vessel to automatically navigate in an unmanned manner to a destination port indicated by the destination information; and a controller configured or programmed to issue an arrival notification and perform a fixed point holding of the marine vessel when the marine vessel arrives at the destination port.

2. The unmanned delivery system according to claim 1, wherein the acquisition unit is configured to acquire correspondence information that associates the cargo with the destination information;

and the controller is configured or programmed to include the correspondence information in the arrival notification.

3. The unmanned delivery system according to claim 1, wherein the automatic navigation unit is configured to cause the marine vessel to depart on a condition that a shipper of the cargo has generated departure permission information permitting departure.

4. The unmanned delivery system according to claim 1, wherein the controller is configured or programmed to guide a recommended loading area of the cargo in the marine vessel based on a weight and/or a size of the cargo.

5. The unmanned delivery system according to claim 1, wherein the automatic navigation unit is configured to start a departure on condition that a total weight of the cargo loaded on the marine vessel is within a first predetermined amount or that a total weight of the marine vessel including the cargo is within a second predetermined amount.

6. The unmanned delivery system according to claim 1, wherein the controller is configured or programmed to transmit the arrival notification to a communication terminal device corresponding to the destination port.

7. The unmanned delivery system according to claim 1, wherein the controller is configured or programmed to function as a transmission unit to transmit, to a communication terminal device corresponding to the destination port, display information to display at least one of a list of cargoes to be unloaded at the destination port, an estimated time of arrival at the destination port, or an estimated time of departure from the destination port.

8. The unmanned delivery system according to claim 1, wherein the controller is configured or programmed to perform a cargo unloading permission process to enable unloading of the cargo in response to acquisition of unloading preparation completion information indicating that preparation to receive the cargo is completed.

9. The unmanned delivery system according to claim 8, wherein the cargo unloading permission process includes unlocking a door of a cargo room or unlocking a door to enter the marine vessel.

10. The unmanned delivery system according to claim 1, wherein the automatic navigation unit is configured to cause the marine vessel to depart on condition that a recipient at a destination of the arrival notification has generated departure permission information permitting departure.

11. The unmanned delivery system according to claim 10, wherein in a case where a predetermined period of time has elapsed after the fixed point holding is started at the destination port, the automatic navigation unit causes the marine vessel to depart even if the departure permission information is not generated.

12. A method for controlling an unmanned delivery system in a marine vessel, the method comprising: acquiring destination information of a cargo to be delivered; causing a marine vessel to automatically navigate in an unmanned manner to a destination port indicated by the destination information; and issuing an arrival notification and performing a fixed point holding of the marine vessel when the marine vessel arrives at the destination port.

13. A marine vessel comprising: the unmanned delivery system according to claim 1.
